Olympic Sports Analysis Report

Structured MECE Framework Breakdown (SQL EDA, Power BI & Dashboard Insights), 1896 to 2016

51	66	757	129K	39K	228
OLYMPIC GAMES	TOTAL SPORTS	TOTAL EVENTS	PARTICIPATIONS	MEDALS	NOCs

1. Executive Summary & Scope

This report provides a Mutually Exclusive, Collectively Exhaustive (MECE) analysis of historical Olympic Games data from 1896 to 2016. It synthesises 18 SQL exploratory data analysis (EDA) queries, 18 Power BI questions answered through a DAX measure layer, and a 5-page interactive dashboard, mapping multi-decade trends across seasonal dynamics, discipline concentration, athlete demographics and national medal dominance.

- **Games:** 29 Summer (56.86%) and 22 Winter (43.14%) editions; Athina and London lead as hosts with 3 Games each.
- **Sports and events:** 66 sports and 757 events; Athletics and Shooting hold 83 events each (10.96% of events each).
- **Participants:** 129K participations, 73.9% male and 26.1% female, with a mean athlete age of 25.78 years.
- **Medals:** 39K medals; the USA leads with 5,414 (2,533 gold). NOCs grew from 11 in 1896 to a peak of 215 in 2012.

2. MECE Framework Taxonomy

PROJECT ANALYTICAL PILLARS (NON-OVERLAPPING & EXHAUSTIVE)

Every analysis question is assigned to exactly one of four pillars, so findings do not overlap and together cover the whole dataset.

- **Pillar I: Games Architecture & Temporal Evolution.** Seasonal distribution (Summer vs. Winter), hosting periodicity across decades, and top host cities.
- **Pillar II: Sports & Event Portfolio Dynamics.** Discipline concentration across 66 sports and 757 events, event expansion over time, and gender category split.
- **Pillar III: Demographic Profile & Participation Trends.** Participation volume (129K), gender balance, age distribution and athlete physical profile.
- **Pillar IV: Geopolitical Dominance & Medal Economics.** NOC participation expansion (228 NOCs), top participating nations, and medal distribution (39K medals).

Core Dataset Baseline Summary

The dataset spans 51 modern Olympic Games from 1896 to 2016: 29 Summer Games (56.86%) and 22 Winter Games (43.14%), 129,000 athlete participations, 39,000 awarded medals (13K gold, 13K silver, 13K bronze) and 228 participating regional NOCs.

3. Pillar I: Games Architecture & Hosting Trends

Summer Games make up the majority of editions, and host recurrence is concentrated in historical European and North American cities. The 1900s decade held the most Games (4, all Summer); from the 1920s onward most decades hold 2 to 3 Games of each season.

Host city	Games hosted	Season and years	Region
London	3	Summer (1908, 1948, 2012)	Western Europe
Athina (Athens)	3	Summer (1896, 1906, 2004)	Southern Europe
Los Angeles	2	Summer (1932, 1984)	North America
Lake Placid	2	Winter (1932, 1980)	North America
Innsbruck	2	Winter (1964, 1976)	Central Europe
Paris / Sankt Moritz / Stockholm	2 each	Summer / Winter mix	Europe

Six cities have hosted the Games twice. Source: SQL Q1, Power BI Q1 to Q3.

4. Pillar II: Sports Portfolio & Event Concentration

The portfolio spans 66 sports and 757 events, with operational scope concentrated in a few flagship sports. Summer has 52 sports and Winter 17; the total (69) exceeds 66 because some sports appear in both seasons. Art Competitions are counted as a sport (29 events) although discontinued.

Sport	Total events	Season	Share of 757 events
Athletics	83	Summer	10.96%
Shooting	83	Summer	10.96%
Swimming	54	Summer	7.13%
Cycling	44	Summer	5.81%
Sailing	37	Summer	4.89%

Concentration. The top five sports hold 301 of 757 events (39.8%).

Gender categories. 458 events (60.50%) are for Men, 211 (27.87%) for Women and 88 (11.62%) are Mixed. SQL Q7 shows event offerings still skew male in Athletics, Cycling and Shooting, while Sailing has distinct mixed events.

Growth. Summer events grew from about 42 (1896) to about 306 (2016) and Winter events from about 17 to about 98 (Power BI Q9). Source: SQL Q4 to Q9, Power BI Q4 to Q9.

5. Pillar III: Demographic Profile & Athlete Dynamics

Analysis of 129,000 athlete participations covers gender balance, age distribution and participation scale.

GENDER SPLIT (129K PARTICIPATIONS)

95K Male (73.9%)

34K Female (26.1%)

AGE PROFILE

25.78 years

Mean athlete age; 18 to 24 is the largest age group, followed by 25 to 30

Participation scale. Participants per Games rose from 0.07K in 1896 to 11.18K in 2016 (Power BI Q12).

Top participating nations. USA (9,225), UK (5,783), France (5,198), Germany (4,763), Canada (4.7K), Italy (4.7K), Japan (4.0K), Sweden (3.8K), Australia (3.7K) and Poland (3.0K).

Physical profile (1960 to 2016). Average height has stayed around 170 to 175 cm and average weight around 65 to 70 kg (SQL Q10). Source: Power BI Q10 to Q12, Dashboard page 3.

6. Pillar IV: Geopolitical Dominance & Medal Economics

A total of 39,000 medals (13K gold, 13K silver, 13K bronze) have been awarded. The 2008 Summer Games had the largest tally (2,048 medals); the Winter peak is 595 in 2014. Athletics (3,892), Swimming (2,978) and Rowing (2,801) lead medals by sport.

NOC	Total participants	Gold	Total medals	Medal conversion rate
United States (USA)	9,225	2,533	5,414	58.69%
Soviet Union (URS)	2,856	1,135	2,658	93.07%
Germany (GER)	4,763	843	2,395	50.28%
United Kingdom (UK)	5,783	658	1,971	34.08%
France (FRA)	5,198	492	1,686	32.44%

Medal conversion rate = total medals divided by distinct participants (SQL Q12). Team events and repeat medals count towards every athlete, so it is an efficiency indicator rather than a win percentage and can approach 100%.

NOC Expansion Trend
Participating NOCs grew from 11 in the 1896 Athens Games to a peak of 215 in 2012 (213 in 2016), with a Summer/Winter zigzag after 1992 as the two seasons alternate.

Additional SQL findings.

- USA (5,414 medals, 35 editions with medals) and UK (1,971, 35 editions) show the most consistent medal performance (Q13).
- Cold-climate nations (Norway, Austria, Canada, Switzerland) dominate Winter medals; warmer, high-population regions lead in Summer (Q17).
- Largest participation growth since the early period: Russia (+1,789), China (+1,650), Australia (+1,164), South Korea (+995) (Q16).

7. Power BI Dashboard Architecture Synthesis

The dashboard has a multi-page interactive structure with global slicers (Gender, Sport Name, Season, Games Year) and standardised KPI cards for drill-down.

DASHBOARD NAVIGATION MODULES

- **Page 1, Olympic Overview:** executive KPIs (51 Games, 66 Sports, 757 Events), Games by Season donut, Games by Decade chart, host cities.
- **Page 2, Sports & Events:** events per sport, sports by season (52 Summer vs. 17 Winter), participation by sport over time.
- **Page 3, Participants:** gender distribution (73.9% M vs. 26.1% F), age group chart, top participating countries.
- **Page 4, Medal Analysis:** total medal KPI cards (13K each), gold medal leaderboard, medals by sport and type, medals by Games year and season.
- **Page 5, Regional Analysis:** NOC growth over time (11 to a peak of 215), participants and medals by region.

8. Question Coverage Map

Each of the 36 analysis questions is assigned to exactly one pillar, which keeps the framework mutually exclusive and shows where coverage is deepest.

Pillar	SQL questions	Power BI questions	Total
I. Games Architecture	Q1, Q2	Q1, Q2, Q3	5
II. Sports & Events	Q4, Q5, Q7, Q8, Q9	Q4 to Q9	11
III. Demographics	Q10	Q10, Q12	3
IV. Geopolitics & Medals	Q3, Q6, Q11 to Q18	Q11, Q13 to Q18	17
Total	**18**	**18**	**36**

Pillar III has the thinnest coverage, so age and gender trends over time are the natural next area for deeper analysis.

9. Strategic Recommendations & Action Plan

Based on the combined SQL and Power BI findings, three initiatives are recommended:

1. Accelerate Gender Parity in High-Volume Disciplines
Women make up 26.1% of participations, and event offerings in Athletics, Cycling and Shooting still skew male (SQL Q7). Expanding women's events in these foundational sports would lift overall balance.

2. Expand Winter Games Global Infrastructure Support
Winter accounts for only 22 of 51 Games and 17 of 66 sports. Targeted support for emerging NOCs in Asia and South America can broaden participation beyond traditional strongholds.

3. Implement Predictive DAX Analytics
Extend the Power BI model with measures for athlete age trajectories and NOC medal prediction ahead of future Games.